23.2_Assisgnment_1275

AI Assisted Coding

Name : K.Vyshnavi

H.no : 2303A51617

Batch : 07

• Students choose a real-world problem to solve with an AI- assisted application (e.g., e-commerce billing system, health monitoring dashboard, student attendance tracker).

• Prepare a project proposal including:

o Objective o Features o

Tools/technologies to be used o

Expected output

Task 1 – Project Proposal

```python
# project_proposal.py

proposal = {
    "title": "AI-Assisted Student Attendance Tracker",
    "objective": "Automatically track and manage student attendance using AI features.",

    "features": [
        "Mark attendance",
        "Store student records",
        "Generate reports",
        "AI-based attendance prediction"
    ],

    "technologies": [
        "Python",
        "Flask (Backend)",
        "SQLite (Database)",
        "HTML/CSS/JS (Frontend)"
    ],

    "expected_output": "A web app that records attendance and predicts future attendance."
}

for key, value in proposal.items():
    print(f"{key.upper()}:\n{value}\n")
```

```
        "HTML/CSS/JS (Frontend)"
    ],

    "expected_output": "A web app that records attendance and predicts future attendance."
}

for key, value in proposal.items():
    print(f"{key.upper()}:\n{value}\n")
```

```
TITLE:
AI-Assisted Student Attendance Tracker

OBJECTIVE:
Automatically track and manage student attendance using AI features.

FEATURES:
['Mark attendance', 'Store student records', 'Generate reports', 'AI-based attendance prediction']

TECHNOLOGIES:
['Python', 'Flask (Backend)', 'SQLite (Database)', 'HTML/CSS/JS (Frontend)']

EXPECTED_OUTPUT:
A web app that records attendance and predicts future attendance.
```

Task 2 – Application Design

• Use AI tools to assist in:

o Designing system architecture diagrams. o Defining database schema (if needed). o Creating wireframes/UI mockups for frontend design.

Task 2 – Application Design

```
# architecture.py

architecture = """
[ User Interface ]
        ↓
[ Flask Backend API ]
        ↓
[ SQLite Database ]
        ↓
[ AI Model (Prediction) ]
"""

print("System Architecture:\n", architecture)
```

```
System Architecture:

[ User Interface ]
        ↓
[ Flask Backend API ]
        ↓
[ SQLite Database ]
        ↓
[ AI Model (Prediction) ]
```

```python
# database.py

import sqlite3

conn = sqlite3.connect("attendance.db")
cursor = conn.cursor()

# Create table
cursor.execute("""
CREATE TABLE IF NOT EXISTS students (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT,
    attendance INTEGER
)
""")

conn.commit()
conn.close()

print("Database created successfully!")
```

```python
html_code =
<!DOCTYPE html>
<html>
<head>
    <title>Attendance Tracker</title>
</head>
<body>
    <h1>Mark Attendance</h1>
    <form method="POST" action="/mark">
        Name: <input type="text" name="name"><br>
        Attendance: <input type="number" name="attendance"><br>
        <button type="submit">Submit</button>
    </form>
</body>
</html>
"""

print(html_code)
```

```
<!DOCTYPE html>
<html>
<head>
    <title>Attendance Tracker</title>
</head>
<body>
    <h1>Mark Attendance</h1>
    <form method="POST" action="/mark">
        Name: <input type="text" name="name"><br>
        Attendance: <input type="number" name="attendance"><br>
        <button type="submit">Submit</button>
```

Task 3 – AI-Assisted Code Development

• Implement the application in phases:

o Frontend: Web/mobile UI with HTML/CSS/JS (or framework).

o Backend: RESTful APIs or server logic. o

Database: Schema design, SQL queries. o Integration:

API calls, authentication, error handling.

Task 3 – AI-Assisted Code Development

```python
# ai_model.py

from sklearn.linear_model import LinearRegression
import numpy as np

# Sample data: days vs attendance
days = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
attendance = np.array([80, 82, 78, 85, 87])

model = LinearRegression()
model.fit(days, attendance)

# Predict future attendance
future_day = np.array([[6]])
prediction = model.predict(future_day)

print("Predicted Attendance:", prediction[0])
```

```
Predicted Attendance: 87.50000000000001
```

Task 4 – Documentation & Code Quality

• Generate automatic documentation and inline comments using

AI.

• Ensure clear README.md for project setup and usage.

• Conduct AI-assisted code review to refine readability and maintainability.

```python
# readme_generator.py

readme = """
# AI Attendance Tracker

## Setup
1. Install dependencies:
   pip install flask sklearn

2. Run the app:
   python app.py

## Features
- Add students
- View attendance
- Predict attendance

## Author
Student Project
"""

with open("README.md", "w") as file:
    file.write(readme)

print("README.md created!")
```

```
README.md created!
```

Task 5 – Testing & Debugging

• Use AI to: o Generate unit tests. o

Identify vulnerabilities/security flaws. o

Suggest optimizations for performance.

Task 6 – Deployment & Presentation

• Deploy application on a cloud/free hosting platform (Heroku, GitHub Pages, Firebase, etc.).

• Prepare a final presentation including:

o Problem statement o Solution

architecture o Live demo o Reflection

on AI assistance and limitations

Expected Outcomes:

• A complete, deployable software application.

• Students gain experience in end-to-end development with AI tools.

• Improved skills in teamwork, documentation, and ethical AI usage.

Task 6 – Deployment (Basic Script)

```
# deployment.py

import os

print("Steps to Deploy:")
print("1. Push code to GitHub")
print("2. Connect to Render / Heroku")
print("3. Deploy Flask app")
print("4. Access live URL")
```

```
Steps to Deploy:
1. Push code to GitHub
2. Connect to Render / Heroku
3. Deploy Flask app
4. Access live URL
```